AO 2 - neidealități

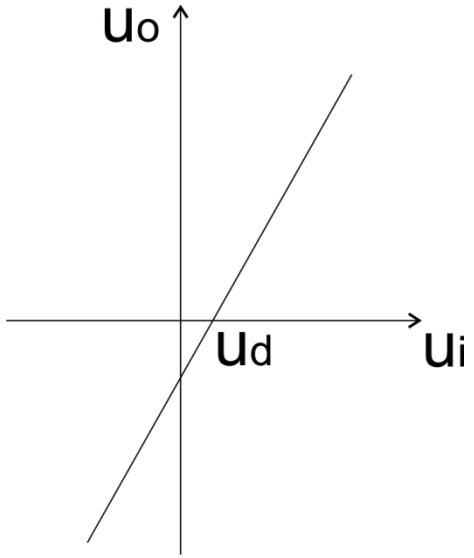
Disclaimer

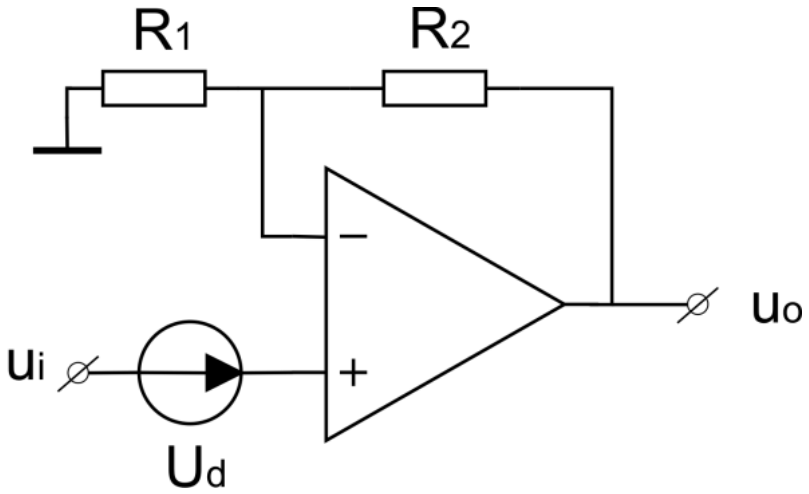
- *Slide-urile acestea reprezintă un rezumat.*
- *Slide-urile nu acoperă toate explicațiile, simulările, demonstrațiile oferite în timpul cursului.*
- *Slide-urile nu limitează în vreun fel materia cerută la examen.*
- *Pentru varianta completă, sunteți așteptați la curs*

Joi 10:00 / 12:00 – on-line, Microsoft teams

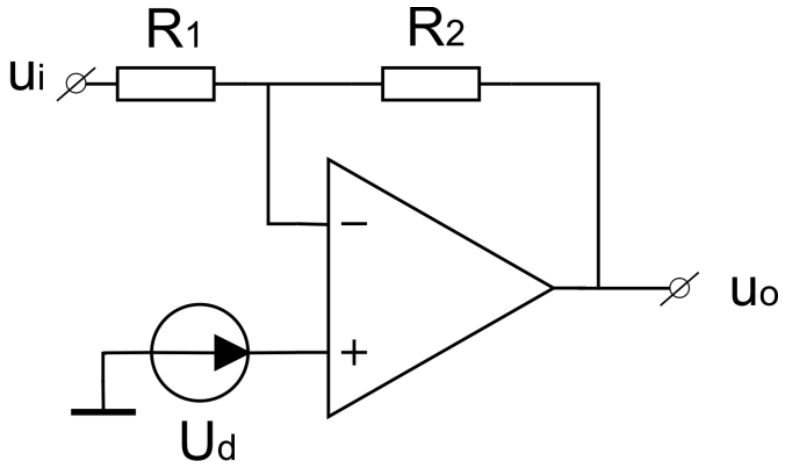
Luni 12:00/14:00, on-line, Microsoft teams

Tensiunea de decalaj Ud (offset voltage)





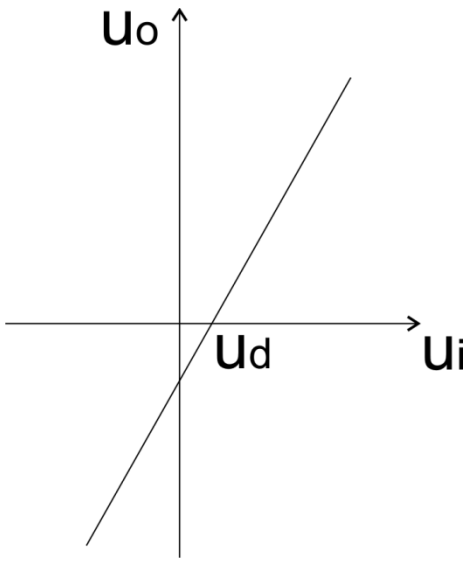
uo = (1+R2/R1)(ui+Ud)

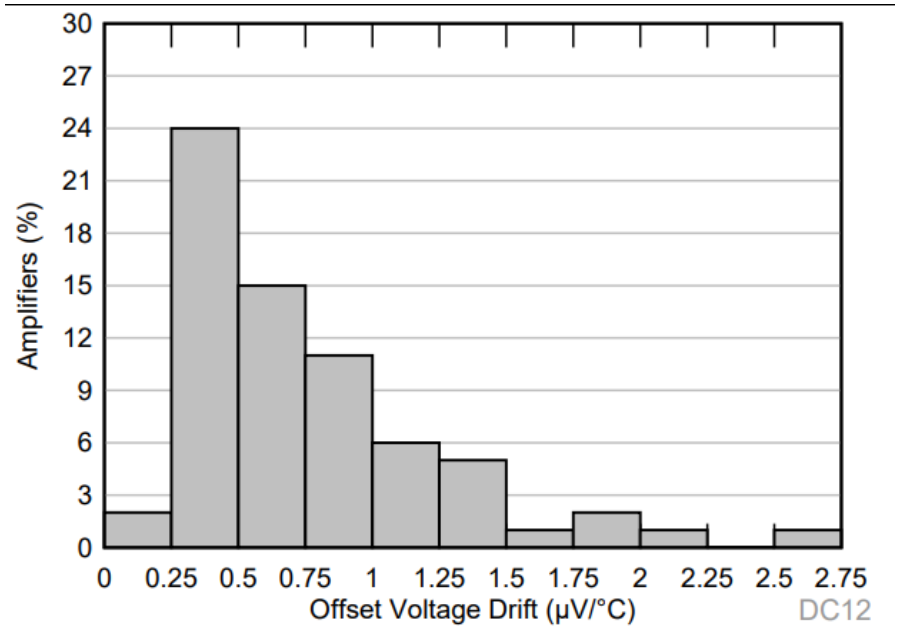


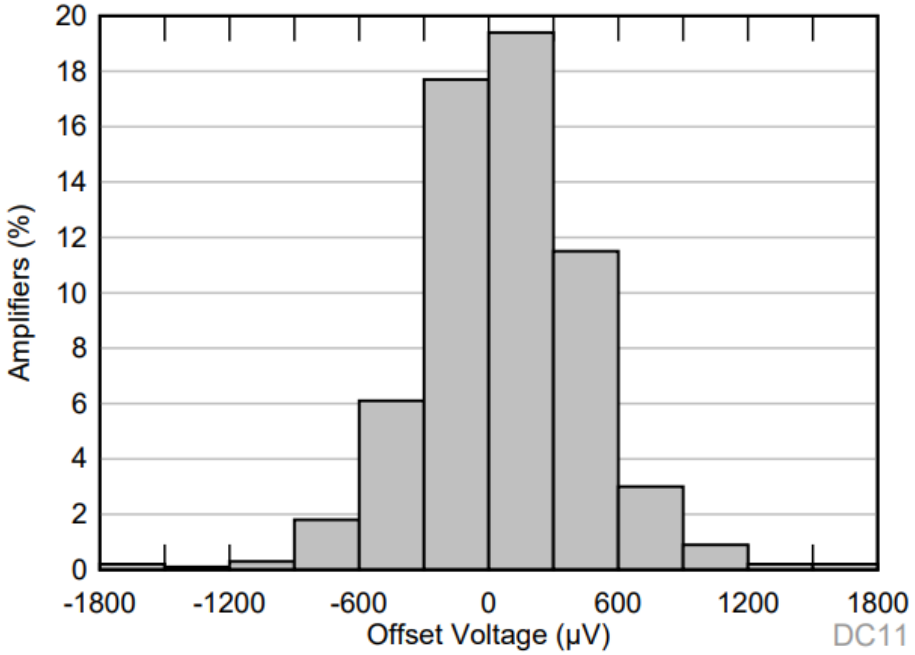
uo = -ui *R2/R1 + Ud(1+R2/R1)

© https://www.ti.com/lit/ds/symlink/lm2904b.pdf?ts=1608108282289

Tensiunea de decalaj Ud (offset voltage)







© https://www.ti.com/lit/ds/symlink/lm2904b.pdf?ts=1608108282289

Curent de polarizare (bias current)

- Ib = (Ib+ + Ib-)/2

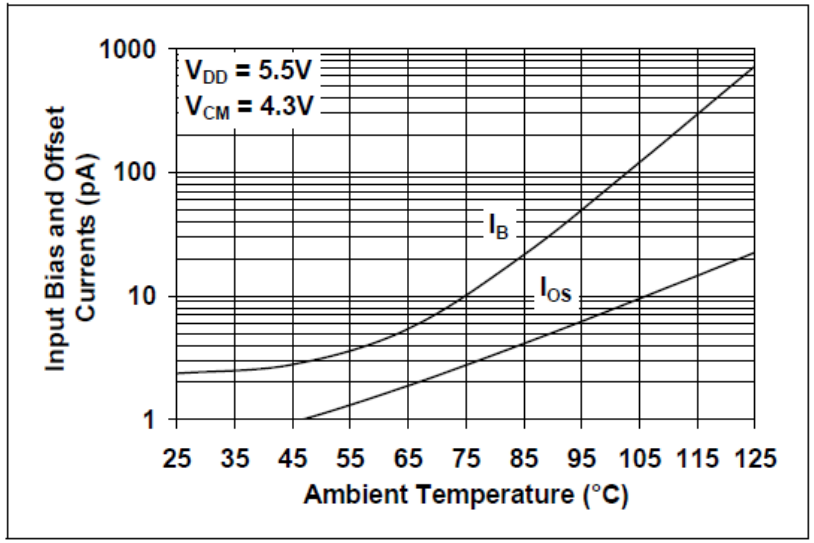
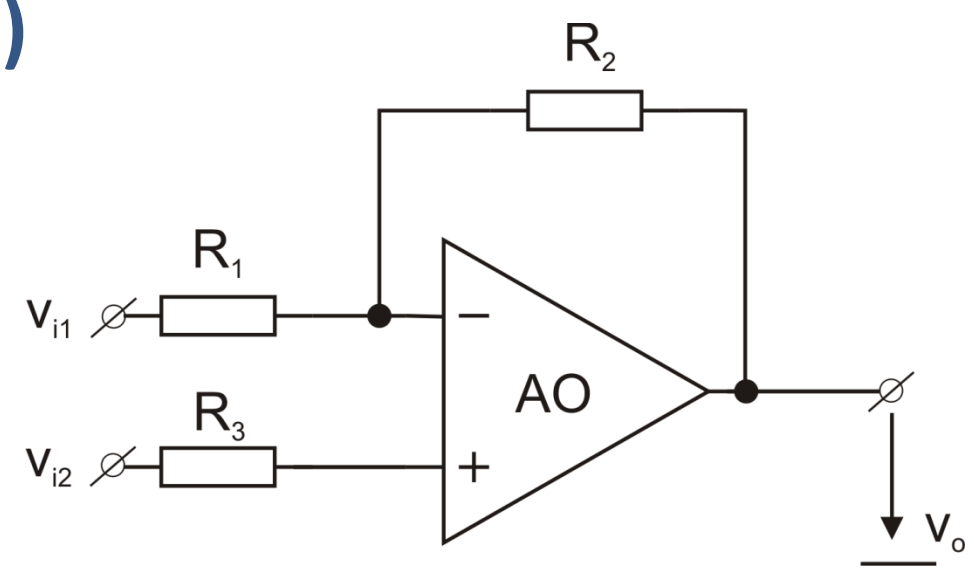


FIGURE 2-14: *Input Bias Current, Input Offset Current vs. Ambient Temperature.*



$$V_{I_b} = I_{b-} R_2 - I_{b+} R_3 \left(1 + \frac{R_2}{R_1}\right)$$

© https://www.ti.com

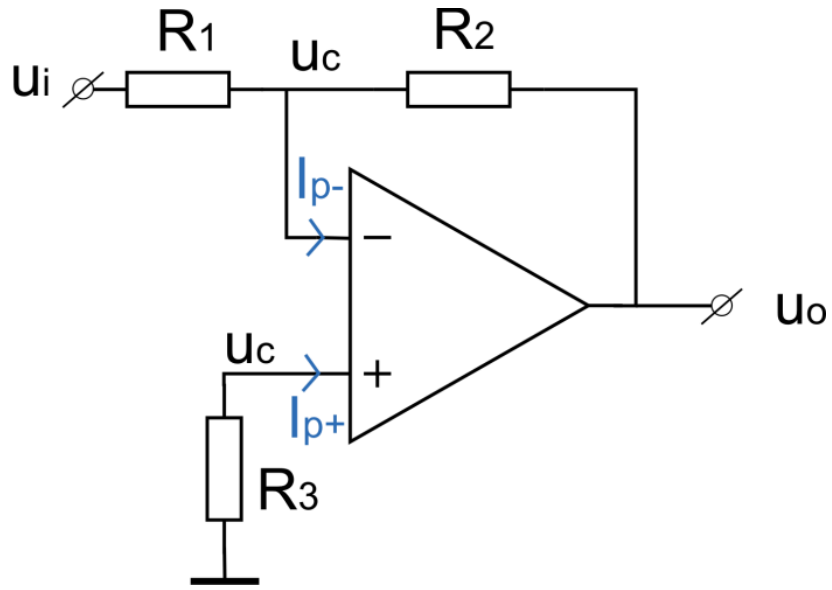
Curent de decalaj (offset current)

- Id = | Ip+ - Ip- |
- uc = -Ip+ R3
- (ui – uc) / R1 = Ip- + (uc-uo)/R2

Ui/R1 + uo/R2 = Ip- + uc(1/R1+1/R2)

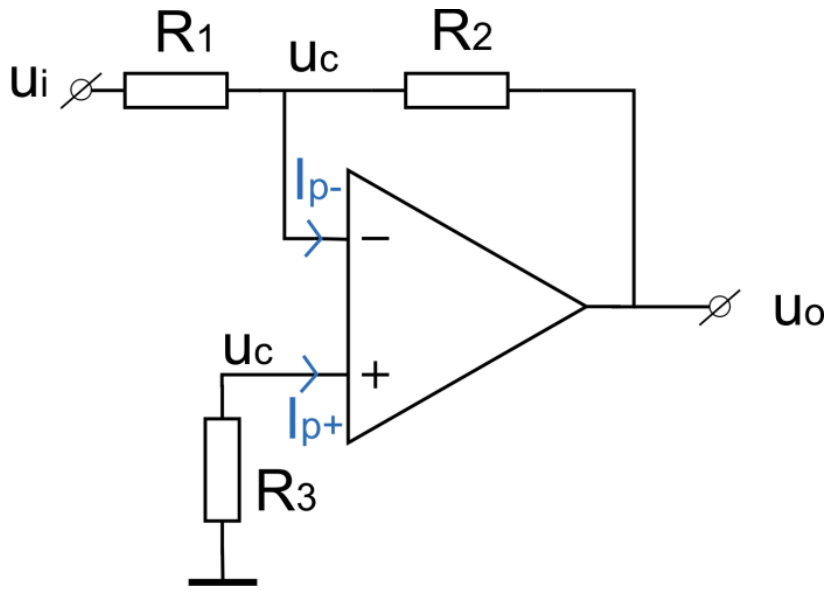
Ui/R1 + uo/R2 = Ip- - Ip+ (R3/(R1||R2)

- Dacă **R3 = R1||R2**
- => ui/R1 + uo/R2 = Id
- => uo = - ui R2/R1 + R2Id



Obs val rez

- preferabil R3 = R1||R2
- 1k – 100k
- Obs val mici:
  - Zgomot mic
  - Consum mare
  - Limitare current
- Obs val mari:
  - Efectul Ib
  - Consum mic
  - Zgomot mare (ex: *Johnson noise)*



Amplificare finită și dependentă de frecvență

© https://www.ti.com

- Exemplu pentru MCP601/1R/2/3/4

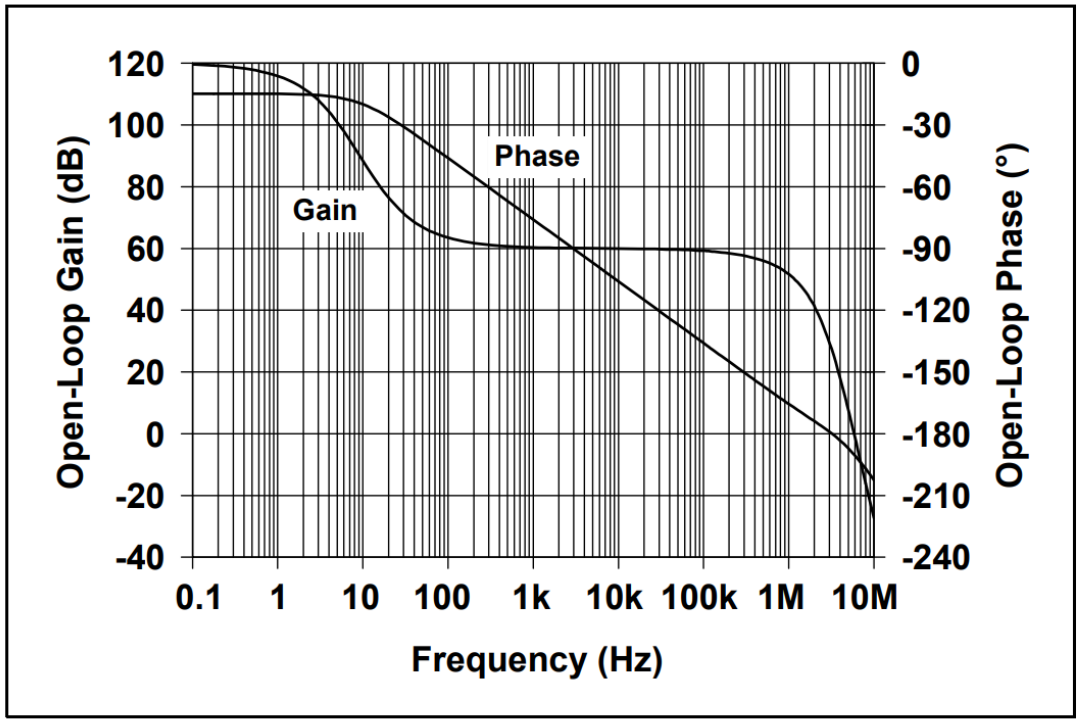


FIGURE 2-1: *Open-Loop Gain, Phase vs. Frequency.*

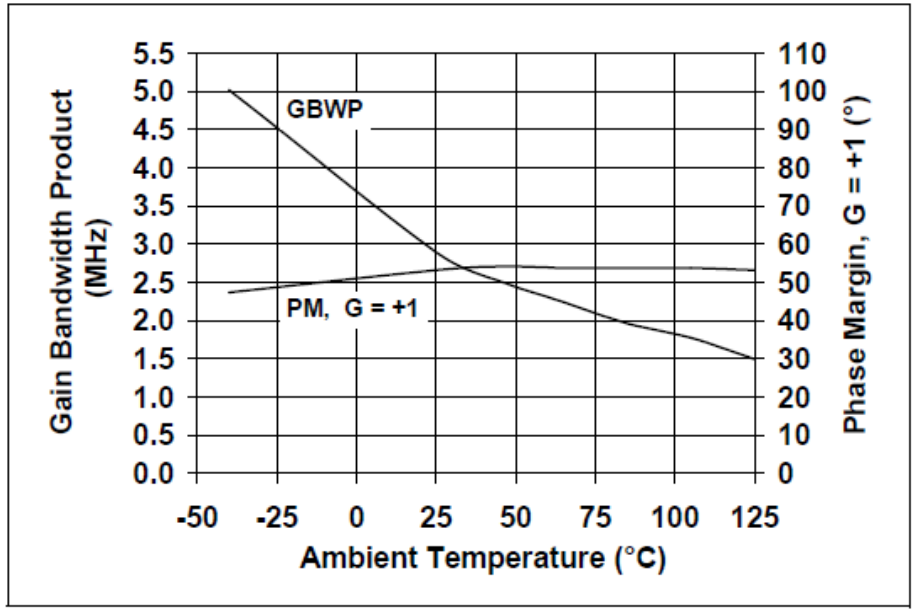


FIGURE 2-3: *Gain Bandwidth Product, Phase Margin vs. Temperature.*

(*Gain-Bandwidth Product*) [kHz / MHz]

Produs amplificare - latime de banda de frecventa

Tensiunea de ieșire finită / limitată

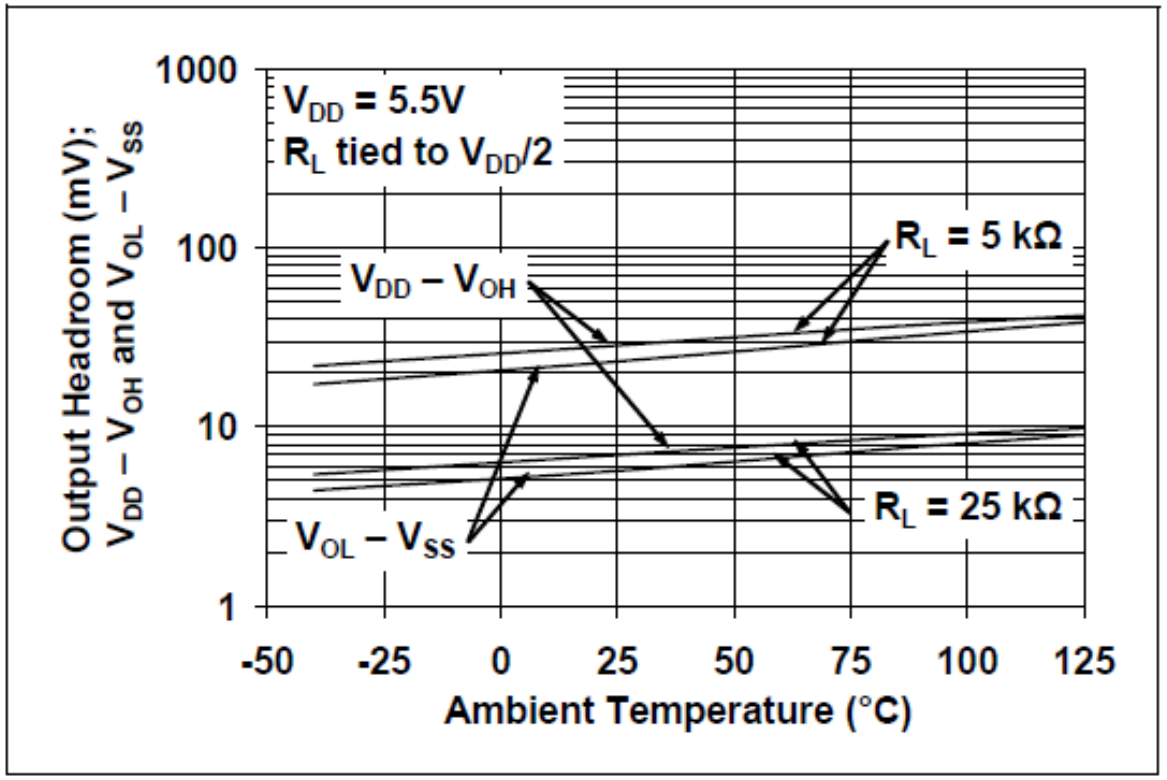


FIGURE 2-23: *Output Voltage Headroom vs. Temperature.*

Rail-to-Rail Input & Output

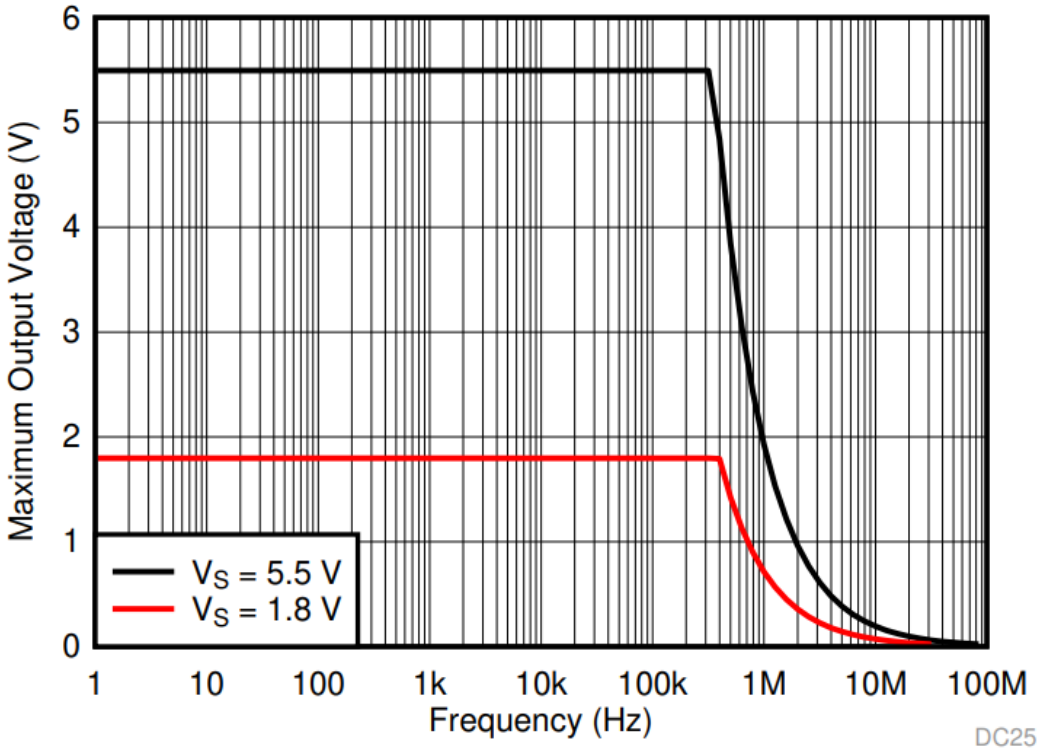
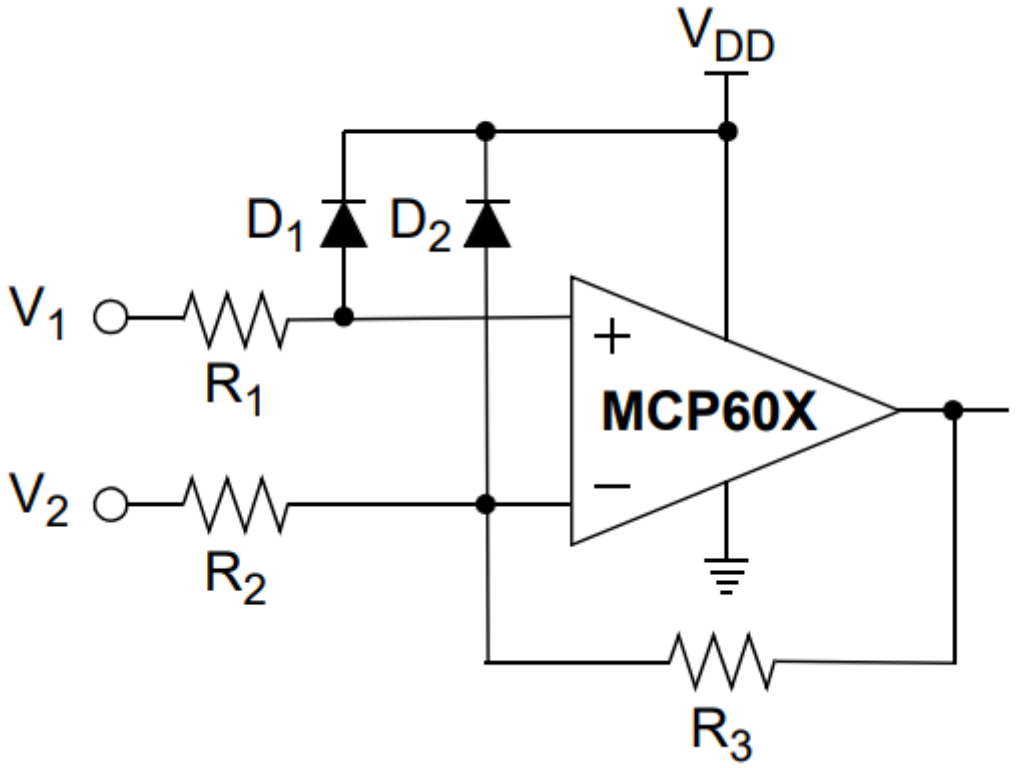


Figure 32. Maximum Output Voltage vs Frequency and Supply Voltage

https://www.ti.com/lit/ds/symlink/tlv9052.pdf

Obs



© https://ww1.microchip.com/downloads/en/DeviceDoc/21314g.pdf

Viteza de variație a tensiunii de ieșire (slew rate)

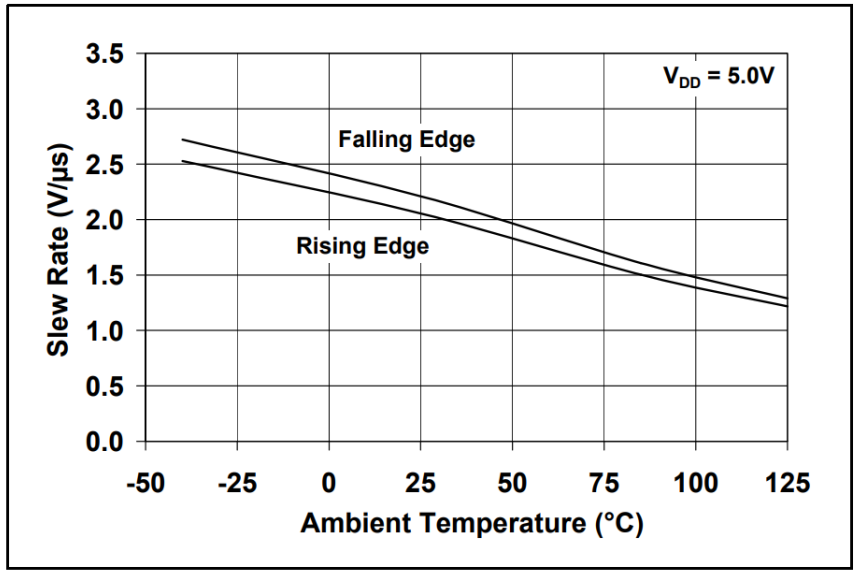
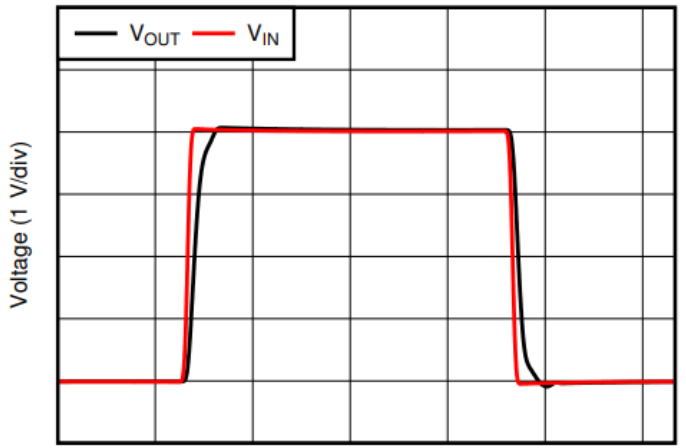
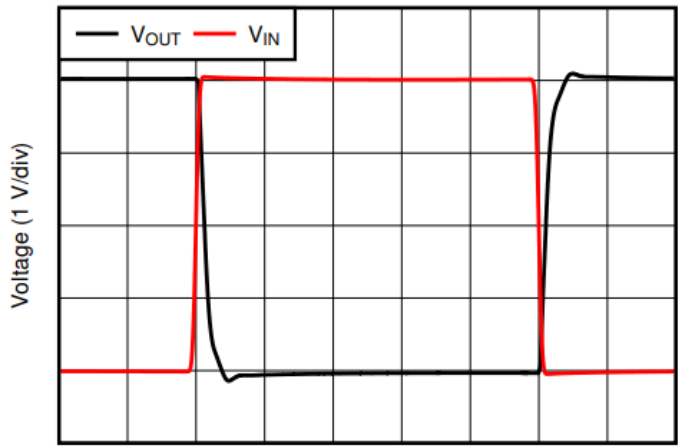


FIGURE 2-2: *Slew Rate vs. Temperature.*



G = +1 V/V
V_{OUT} step = 4 V_{p-p}

Figure 25. Large-Signal Step Response



G = –1 V/V
V_{OUT} step = 4 V_{p-p}

Figure 26. Large-Signal Step Response

Gain limitat funcție de sarcină

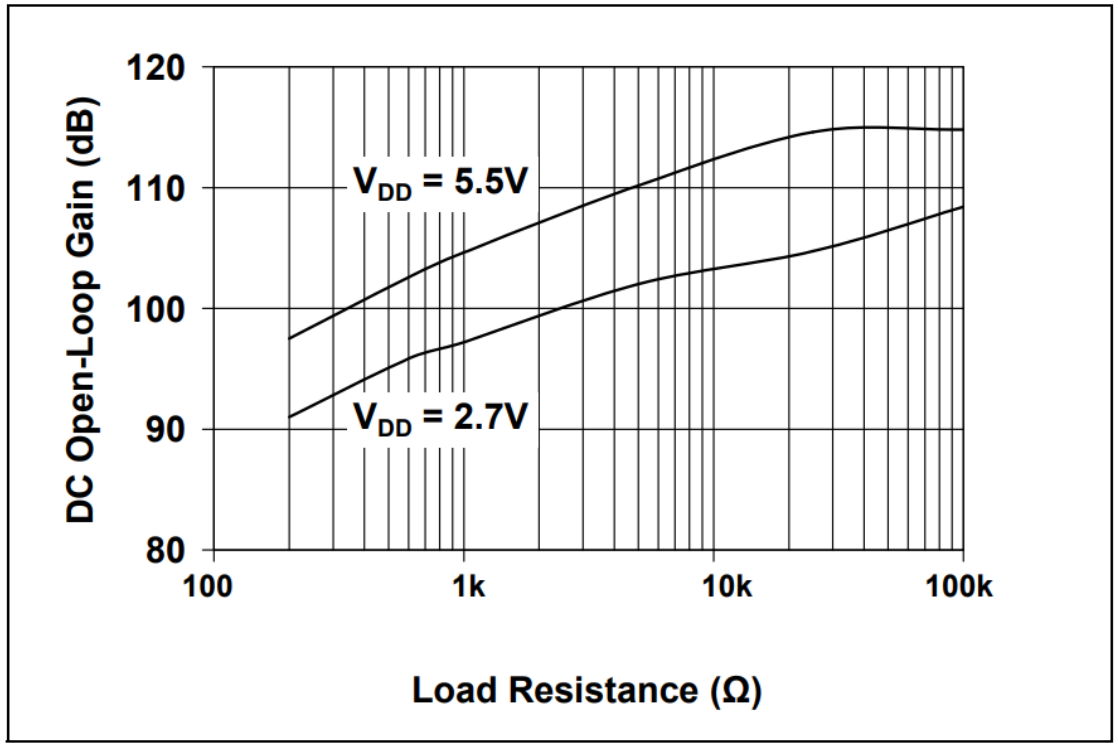
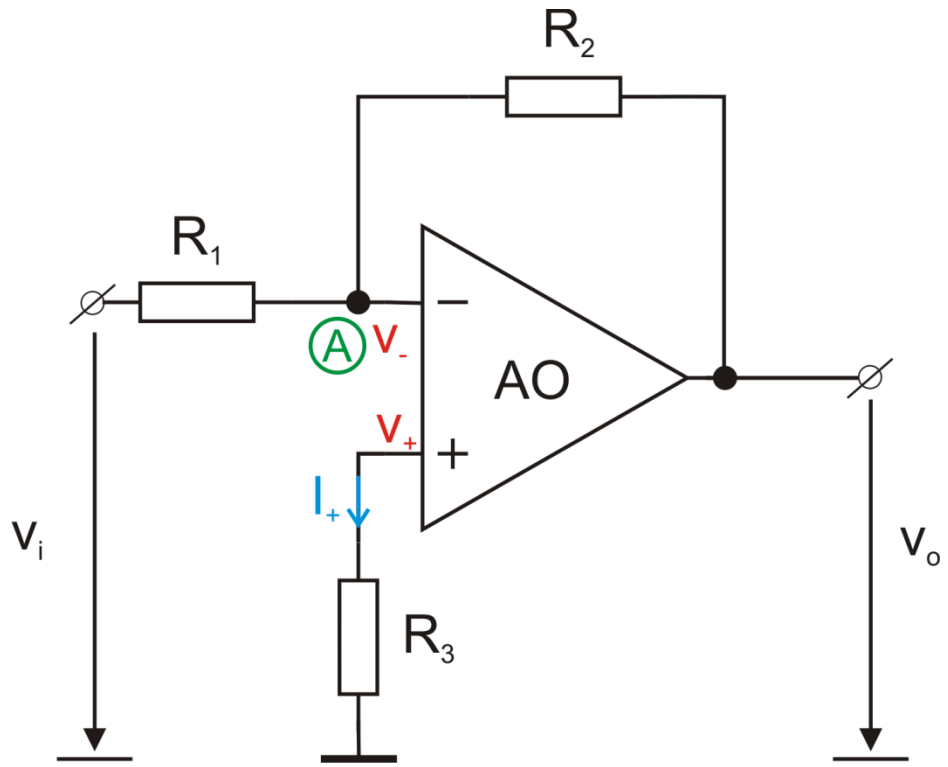


FIGURE 2-15: *DC Open-Loop Gain vs. Load Resistance.*

© https://ww1.microchip.com/downloads/en/DeviceDoc/21314g.pdf

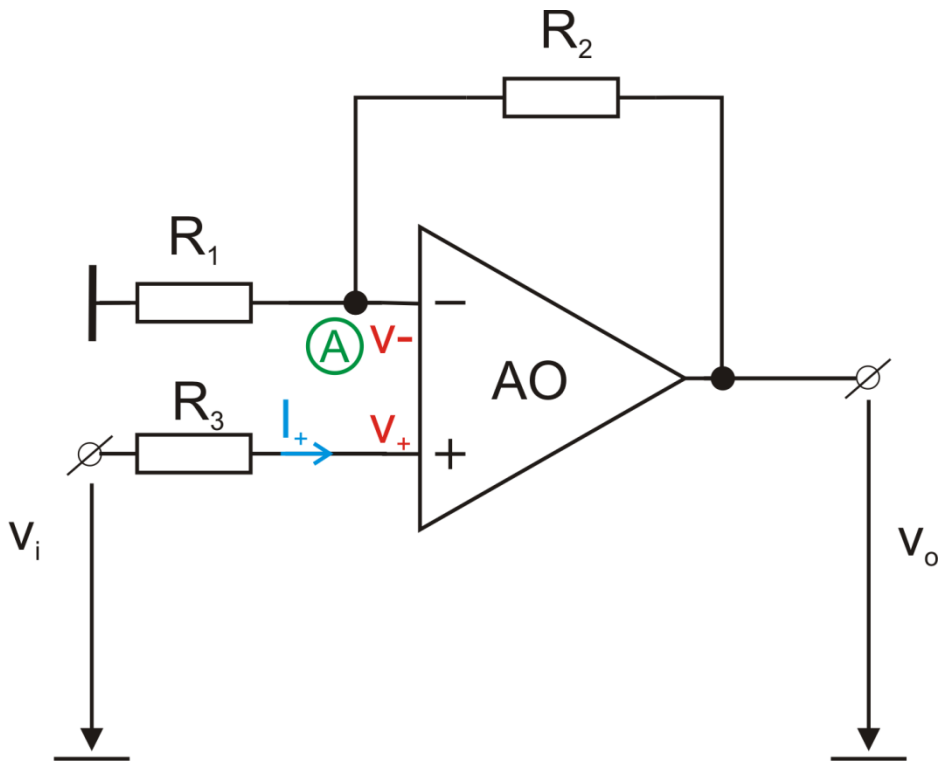
Discutie – amplif neinversor

$$V_{out} = V_{in}\left(1 + \frac{R_2}{R_1}\right)$$



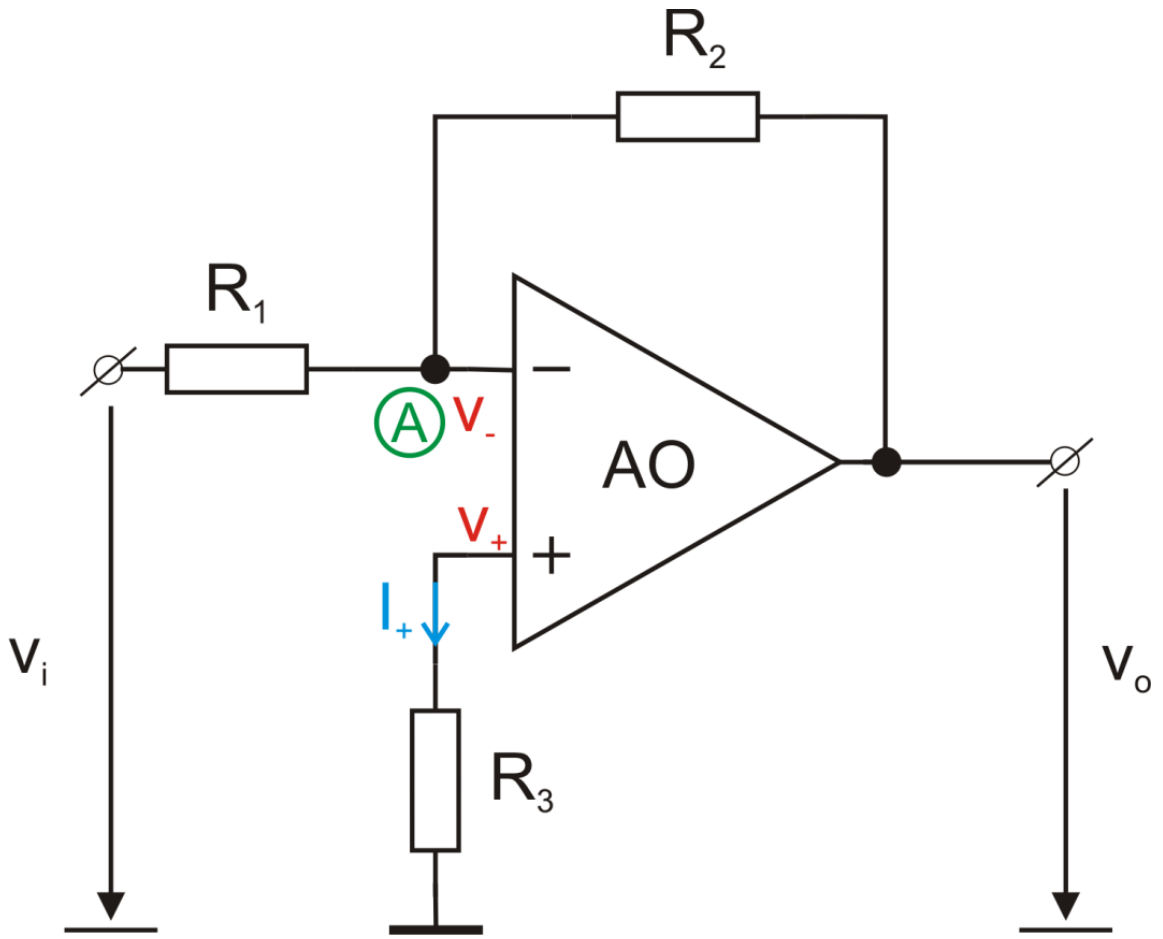
Discutie – amplif neinversor

$$V_{out} = V_{in}\left(1+\frac{R_2}{R_1}\right) + V_{offset}\left(1+\frac{R_2}{R_1}\right) - I_{bias+}R_3\left(1+\frac{R_2}{R_1}\right) + I_{bias-}R_2$$



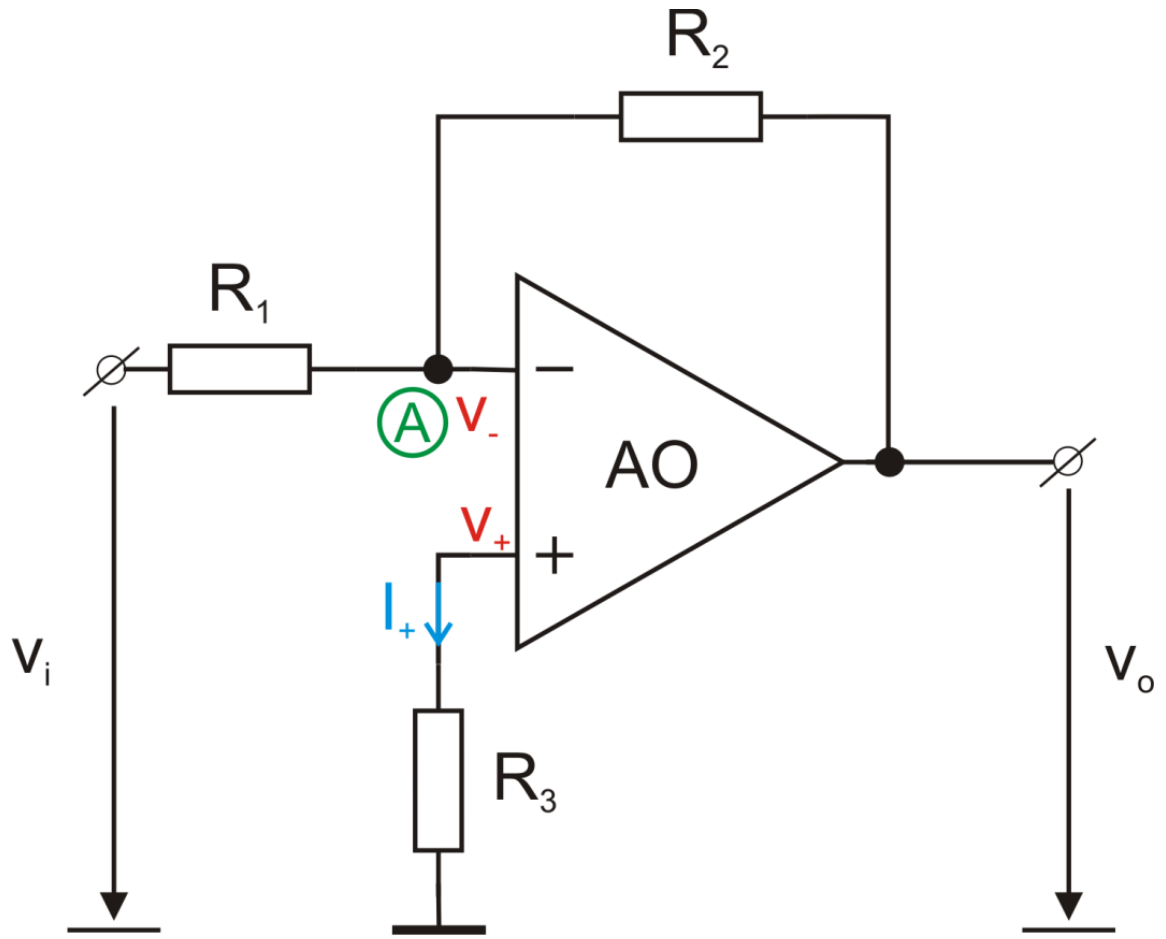
Obs: Neinversorul prezinta o impedanta de intrare foarte mare => comportamentul slab influenta de impedanta de iesire a sursei de semnal & curent slab tras de la sursa de semnal

Discutie – amplif inversor



Discutie – amplif inversor

$$V_{out} = -V_{in}\left(\frac{R_2}{R_1}\right)$$



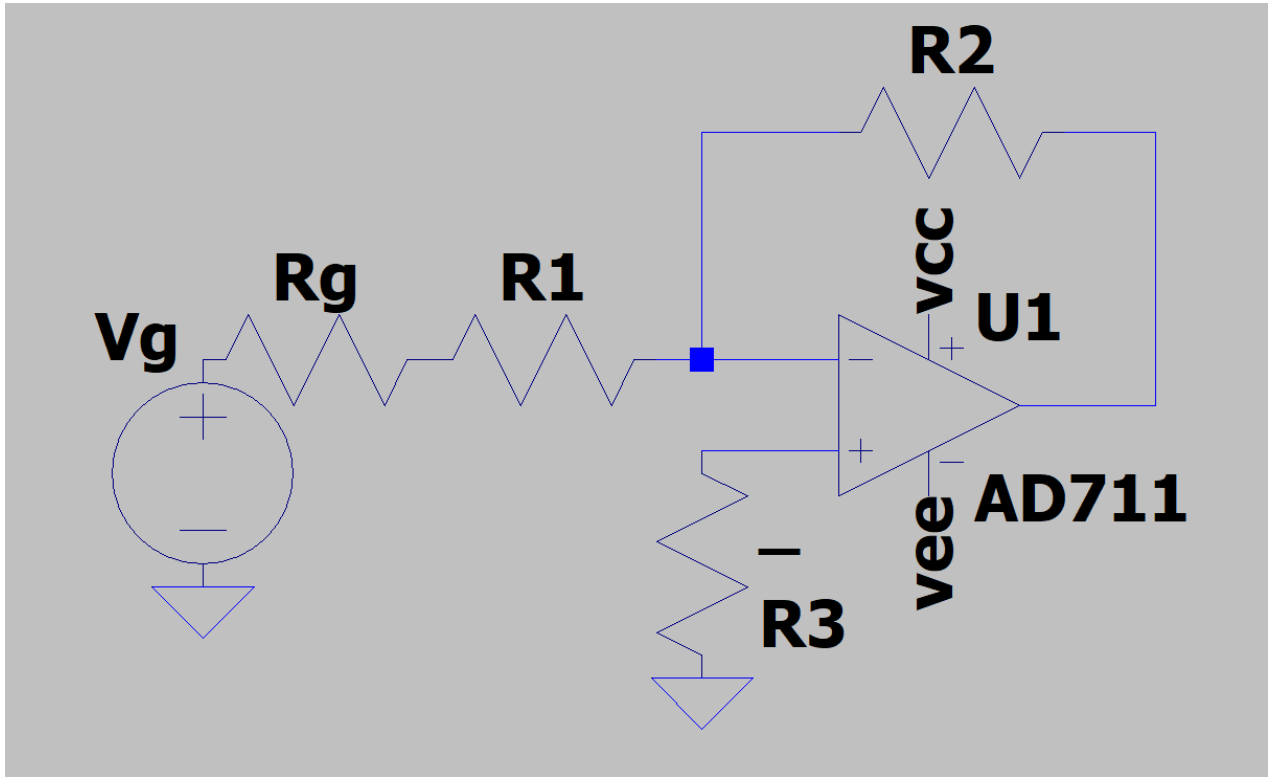
Discutie – amplif inversor

$$V_{out} = -V_g \left(\frac{R_1}{R_1 + R_g}\right)\left(\frac{R_2}{R_1}\right)$$

$$+ V_{offset} \left(1 + \frac{R_2}{R_1 + R_g}\right)$$

$$- I_{bias+} R_3 \left(1 + \frac{R_2}{R_1 + R_g}\right)$$

$$+ I_{bias-} R_2$$



Ex OPAMPS

- LM741 / RC4558
- LM324
- LM358
- LMV358
- TL071
- NE5532
- OP007
- OPA277
- OPA192